

Techniques to Determine Malicious Activity (CS0-003)

Introduction

Objective

Welcome to the **Techniques to Determine Malicious Activity** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

As a Cyber Security Analyst, it is important to constantly monitor the environment for malicious activity and be proactive to advert threats and exploits.

In this module, different techniques will be explored for detecting malicious activity.

Overview

Learning Outcomes:

In this module, you will complete the following exercises:

- Exercise 1 – Monitor Windows Event Log with a PowerShell Script
- Exercise 2 – Monitor Login Events on Linux Devices

After completing this module, you should be able to:

- Enable Audit Logon Events
- Generate Logs for Evaluation
- Create a Logon Events Script using PowerShell ISE
- Prevent a Brute-Force Attack
- Monitor Login Events on a Linux Device

Exam Objectives:

The following exam objectives are covered in this module:

1.3 Given a scenario, use appropriate tools or techniques to determine malicious activity.

- Common techniques

Exercise 1 – Monitor Windows Event Log with a PowerShell Script

A Cyber Security Analyst must constantly monitor the networking environment for malicious activities. Monitoring log files on a Windows system is one of the most common ways to detect environmental anomalies, but it can be repetitive.

In this exercise, a Powershell script will be created that retrieves specific Windows Event log files that might indicate malicious activity.

Learning Outcomes:

After completing this exercise, you should be able to:

- Enable Audit Logon Events
- Generate Logs for Evaluation
- Create a Logon Events Script using PowerShell ISE
- Prevent a Brute-Force Attack

Your Devices:

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server
- **ACIWIN11** – Windows 11 – Domain Member Workstation

Task 1 – Enable Audit Logon Events

Failed log-in events on the devices on the network may indicate a possible brute-force attack on the network. A Cyber Security Analyst can enable the monitoring of log-in attempts to detect a possible brute force attack.

In this task, logon audit events will be enabled to log these events.

1

Click the **Start** button.



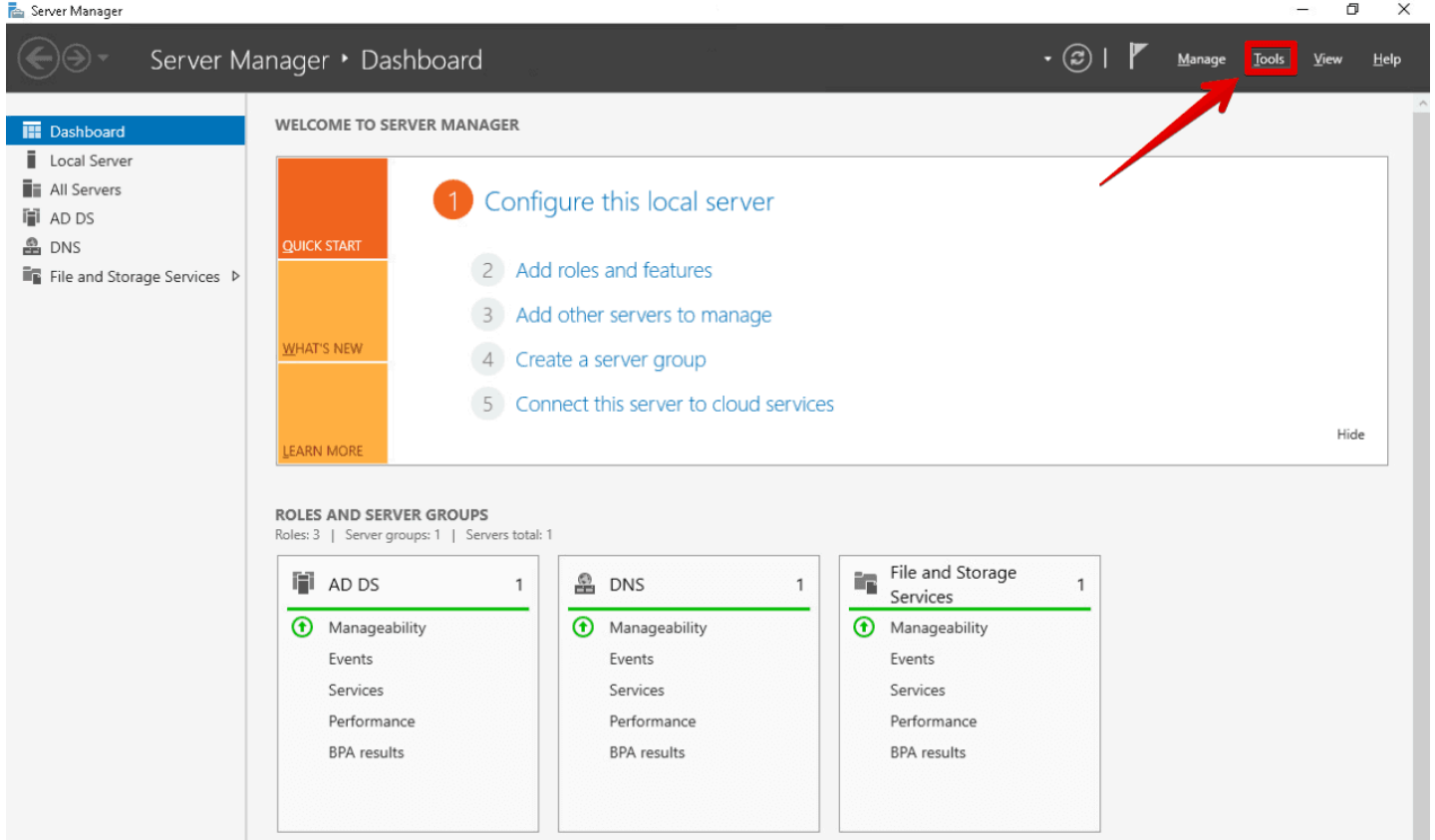
2 **Connect** to **ACIDC01**.

Click **Send+Ctrl+Alt+Delete button** on the top of the screen.

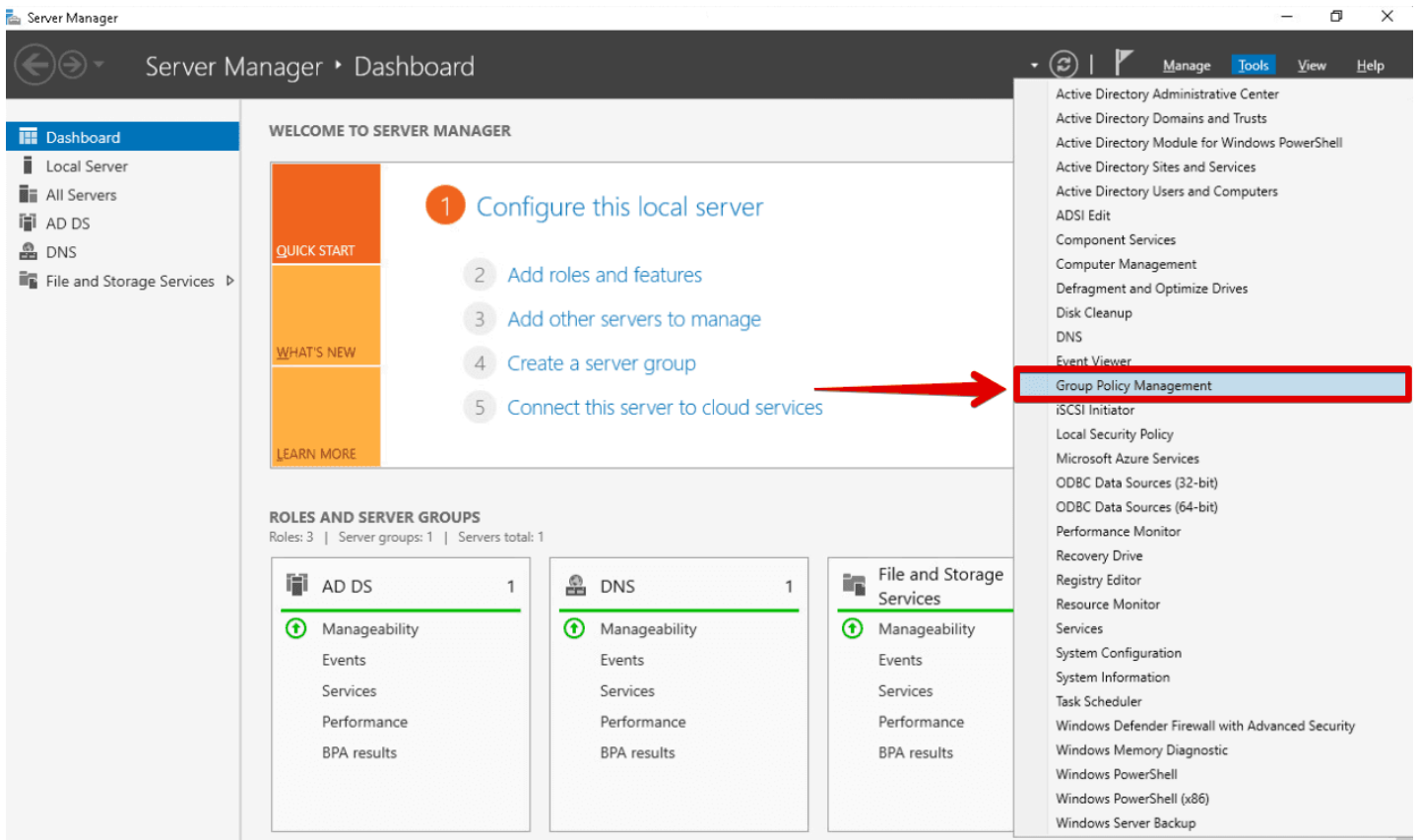
Username: **ACIPLAB\Administrator**

Password:

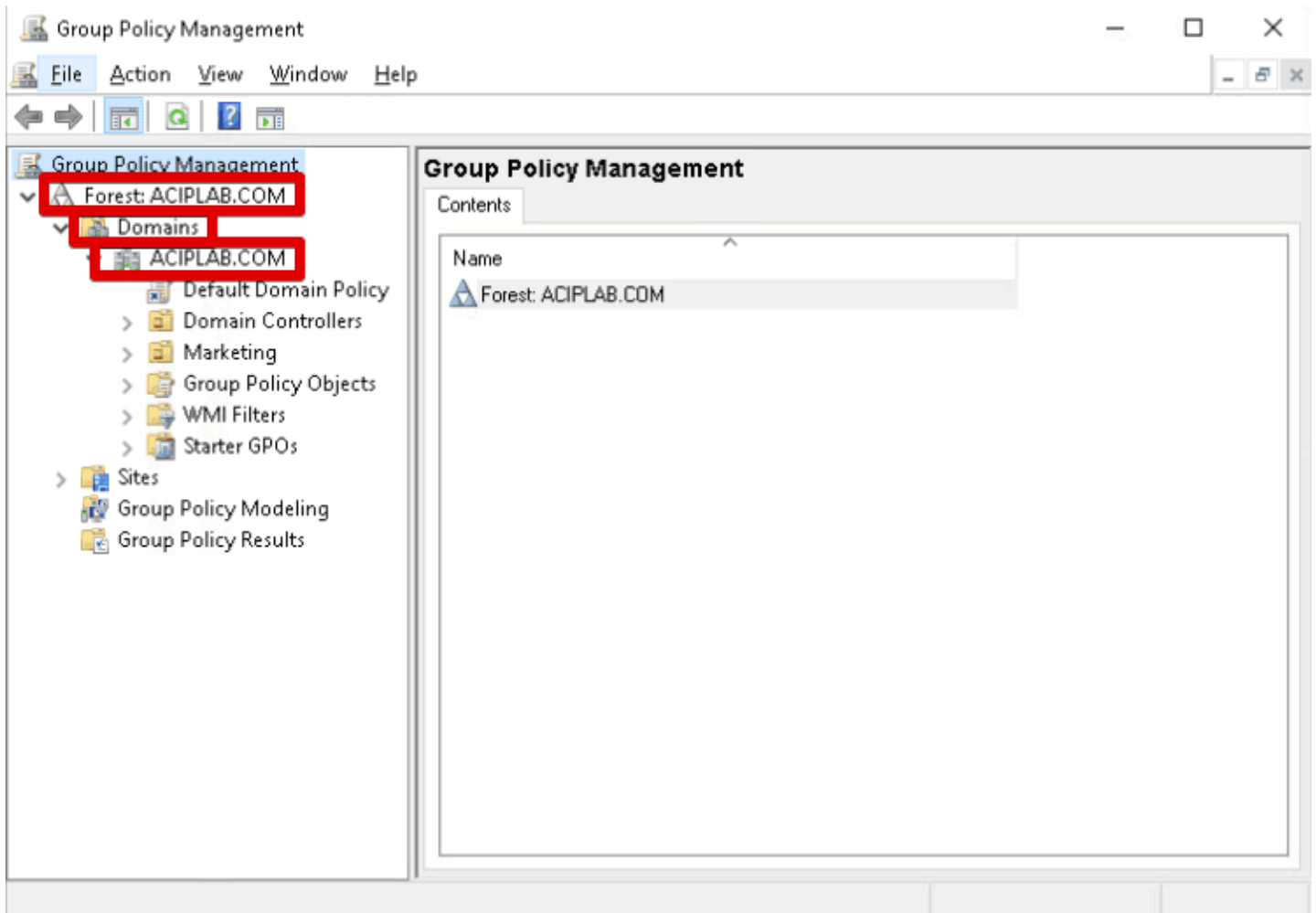
In **Server Manager**, **click** **Tools**.



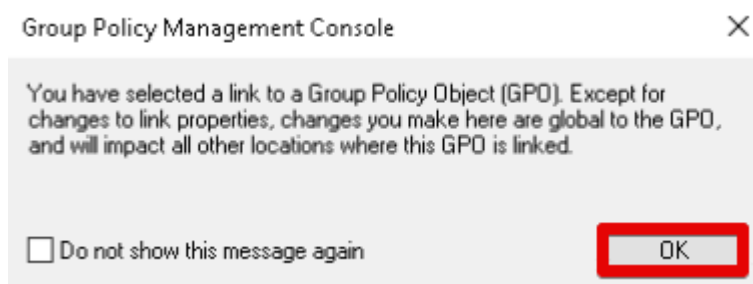
3 In the **Tools** menu, **select** Group Policy Management.



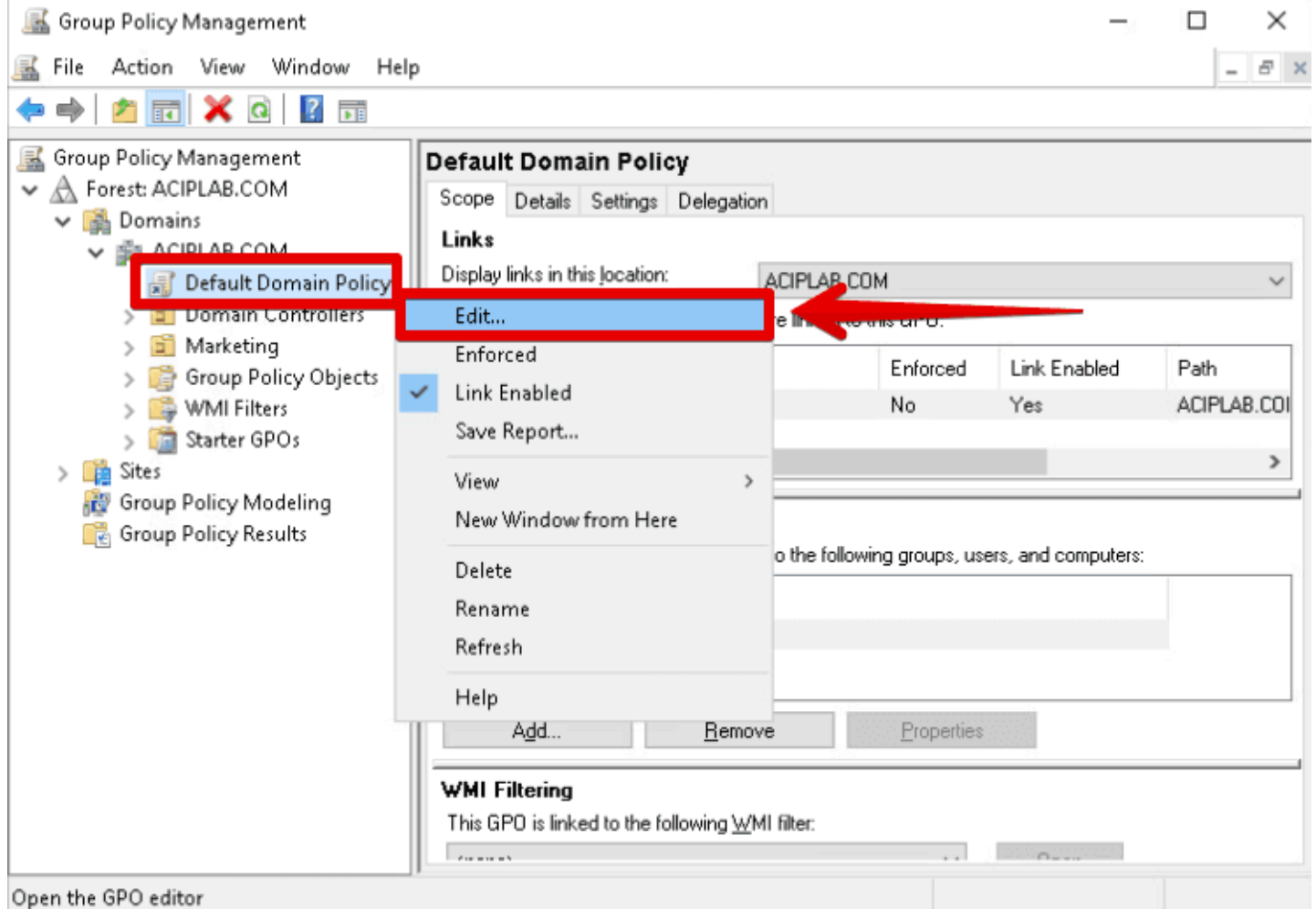
- 4 In the **Group Policy Management** window, **expand** **Forest: ACIPLAB.COM >> Domains >> ACIPLAB.COM**.



- 5 **Select** the **Default Domain Policy** and **click OK** on the pop-up window.



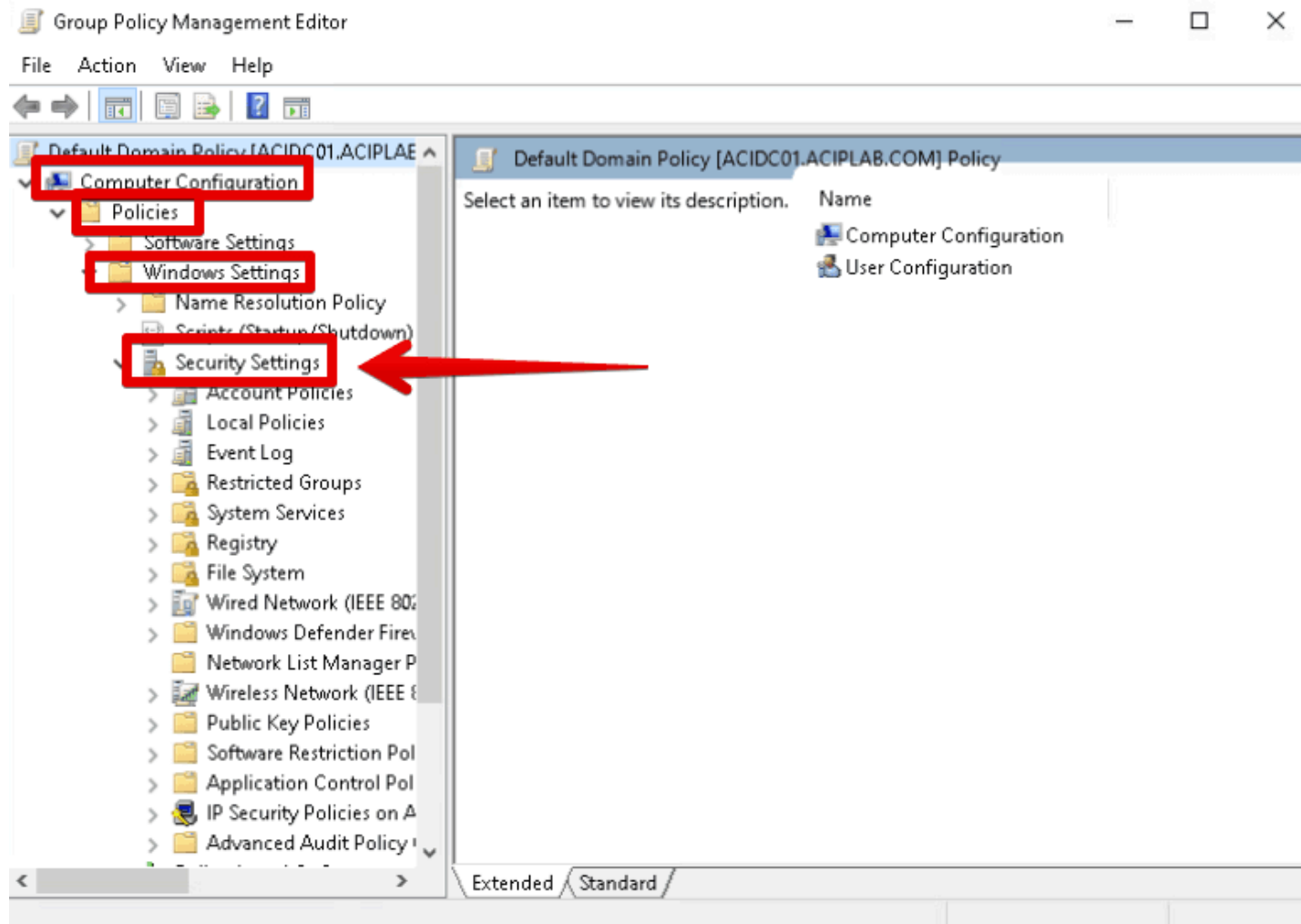
- 6 **Right-click** the **Default Domain Policy** and **select Edit**.



7

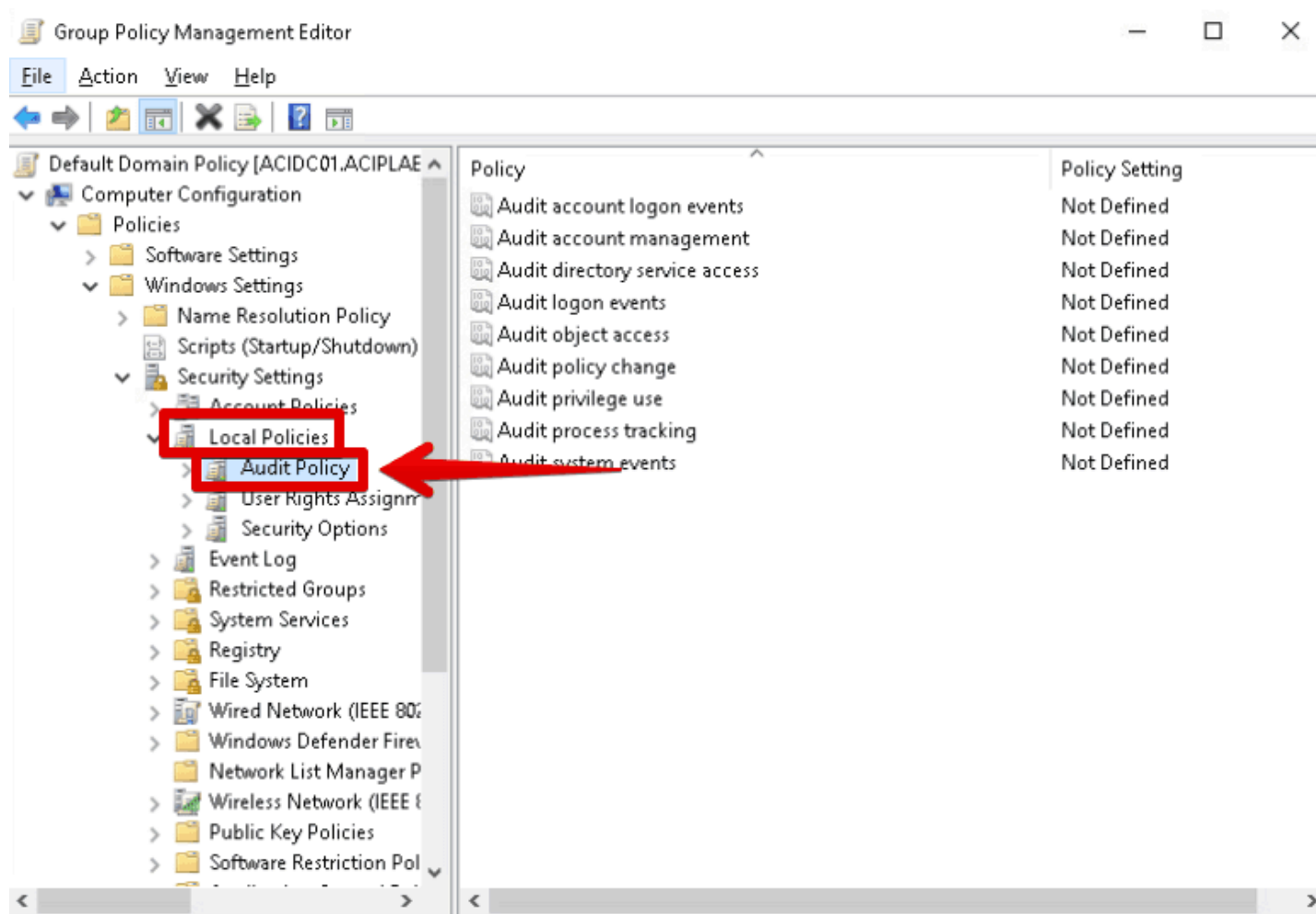
In the **Group Policy Management Editor**, **select** Computer Configuration.

Expand Policies >> Windows Settings >> Security Settings.



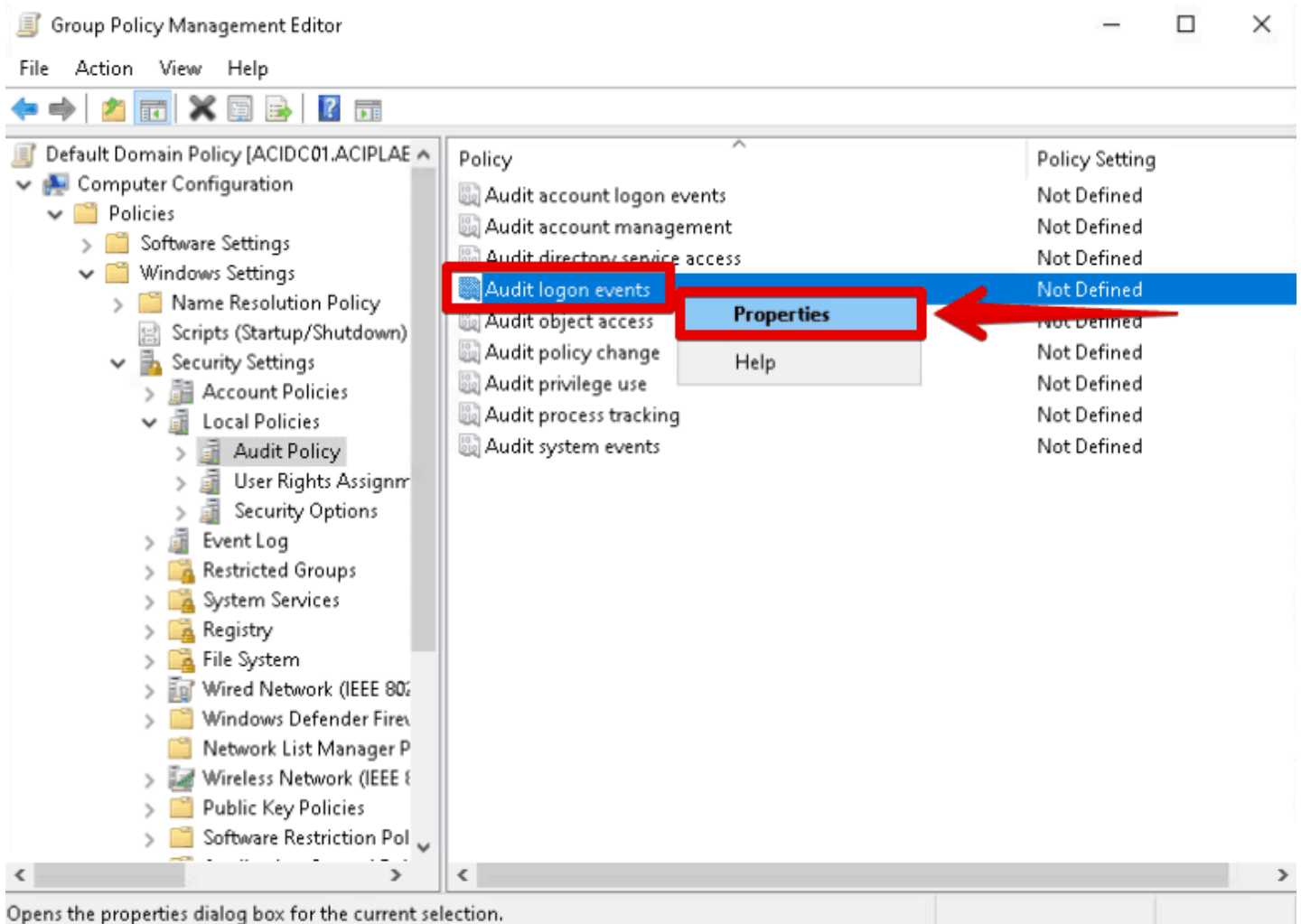
8

In the **Security Settings** pane, **expand** Local Policies and **select** Audit Policy.



9

Right-click **Audit logon events** and **select** **Properties** in the right pane of the **Audit Policy** menu.



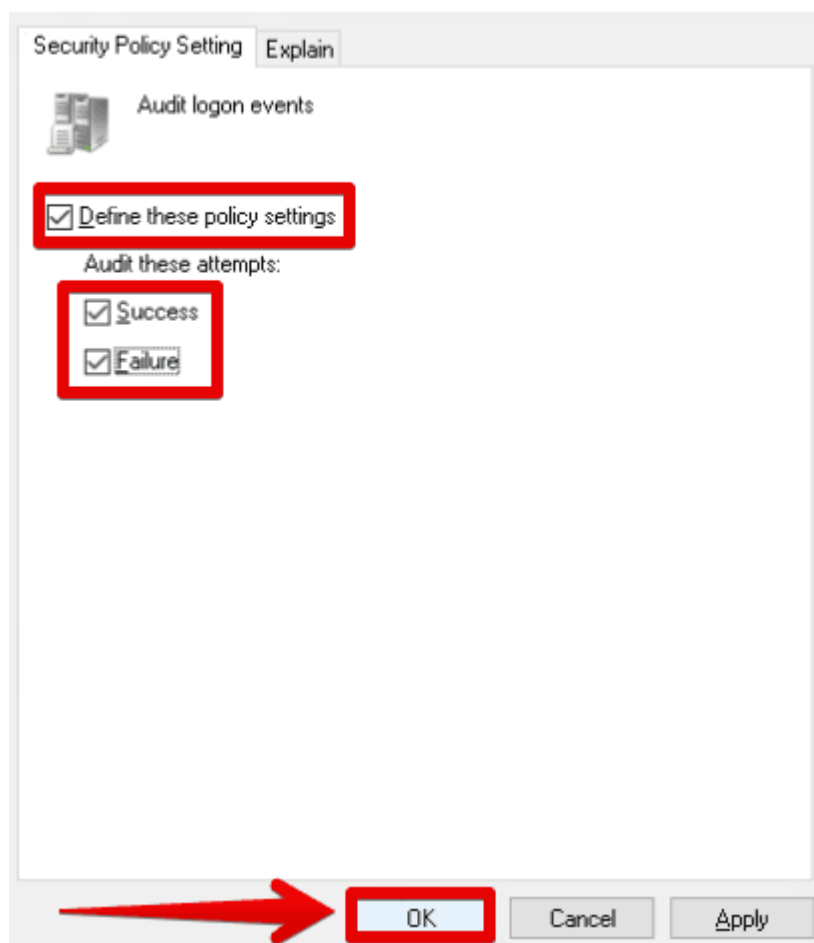
10 In the **Audit logon events** **Properties** pop-up window, **enable** the following tick-boxes:

Define the policy settings

Success

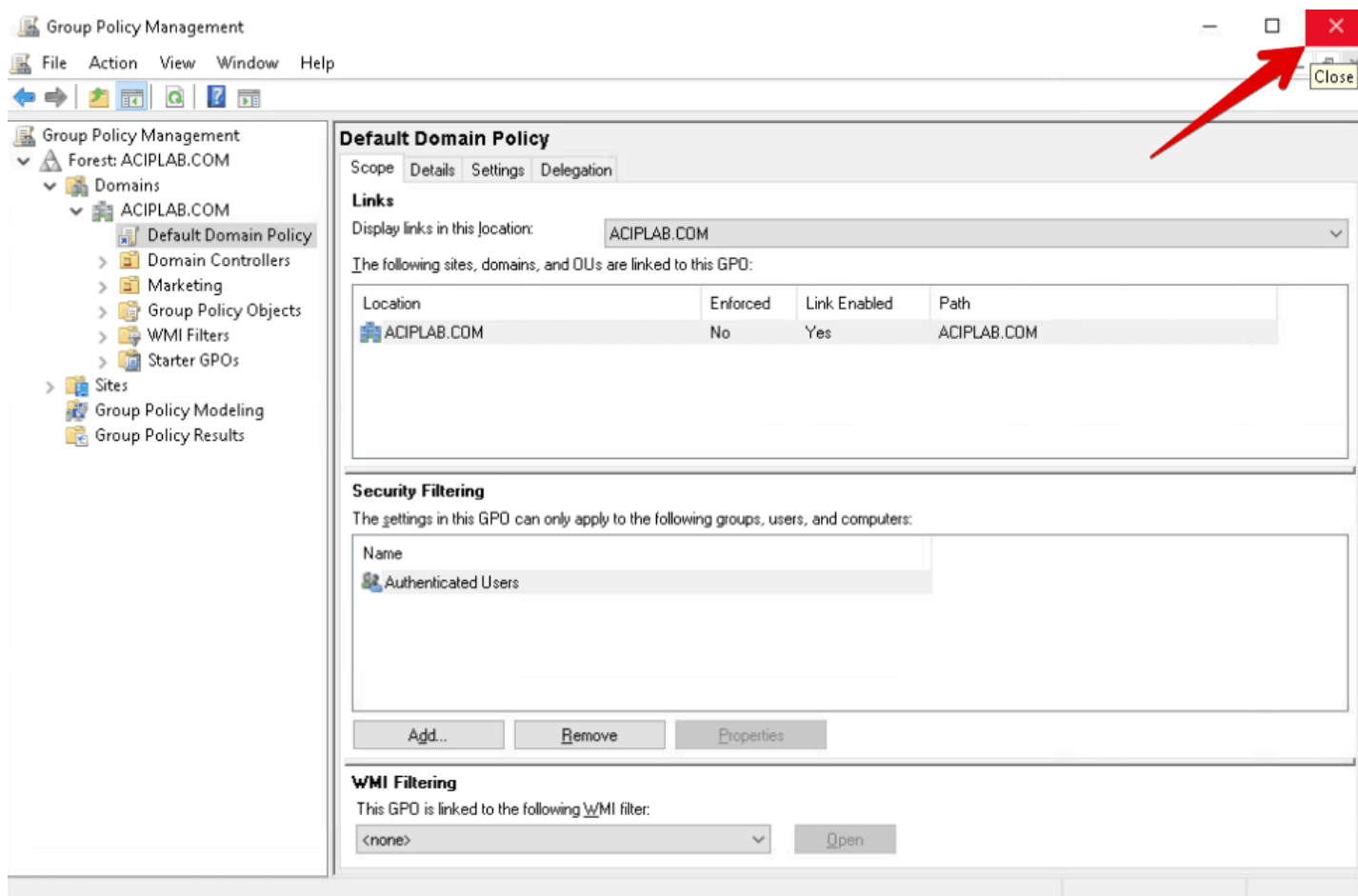
Failure

Click **OK**.



11

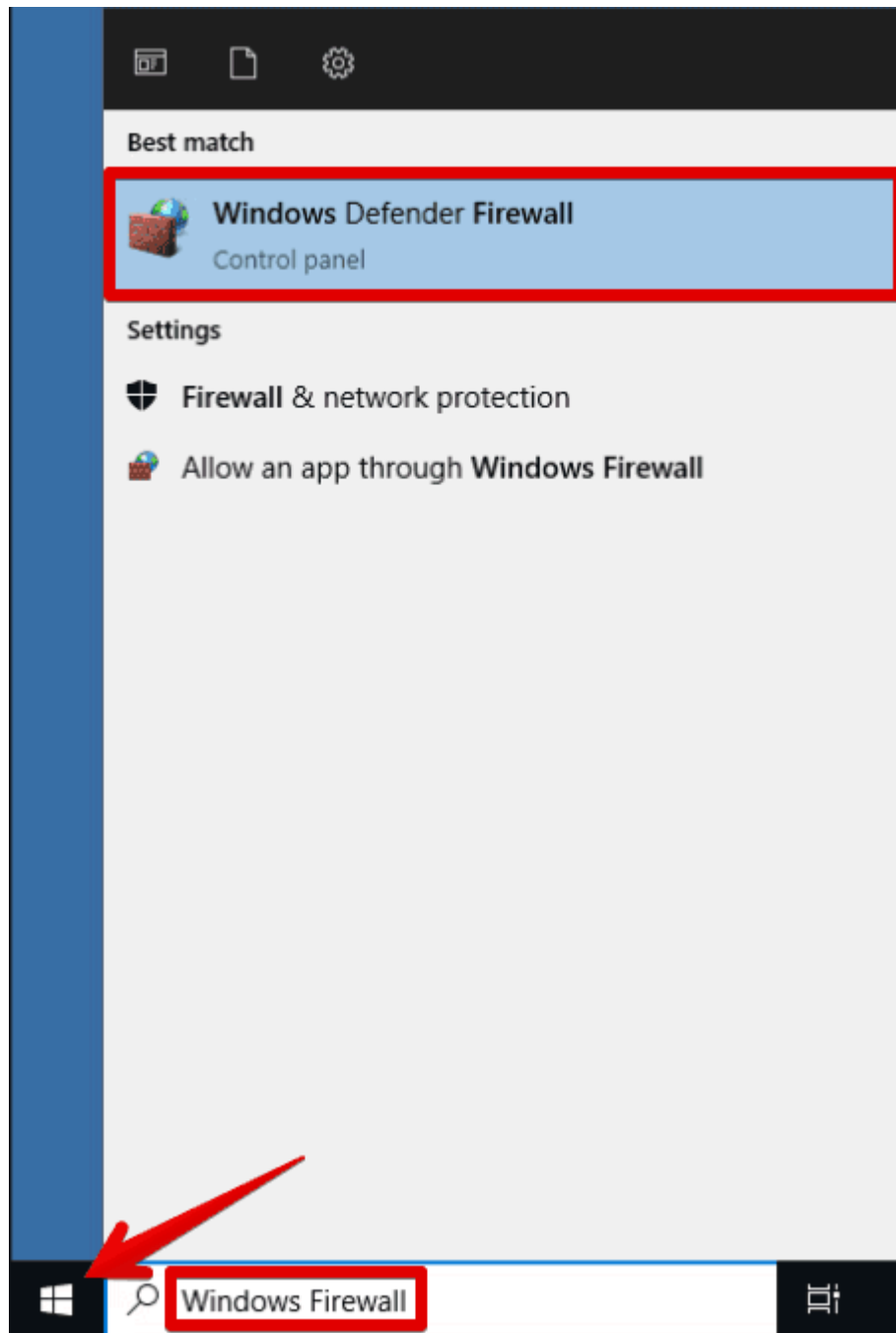
Close the Group Policy Management window.



12

Connect to **ACIDM01**, **click** **Start**, **type** the **following** and **press** **Enter**:

Windows Firewall



13

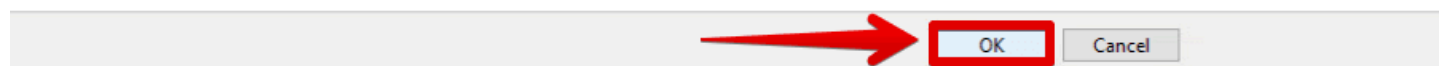
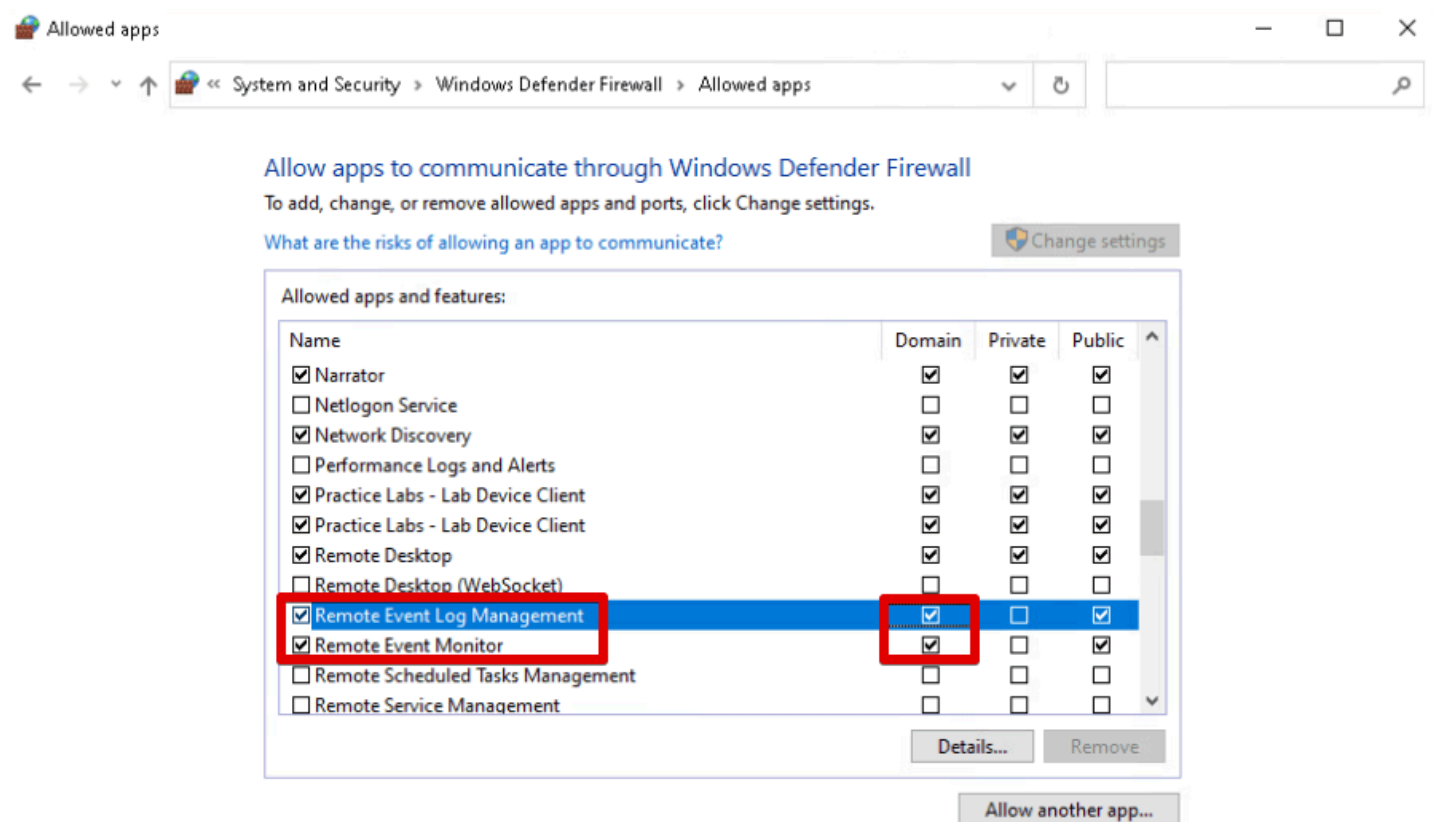
In the **Windows Defender Firewall** window, **click** **Allow an app or feature through** **Windows Defender Firewall**.

14

On the **Allowed apps window**, **scroll down** and **enable** the **Remote Event Monitor** and **Remote Event Log Management** tick boxes.

Ensure the Domain tick-box is selected

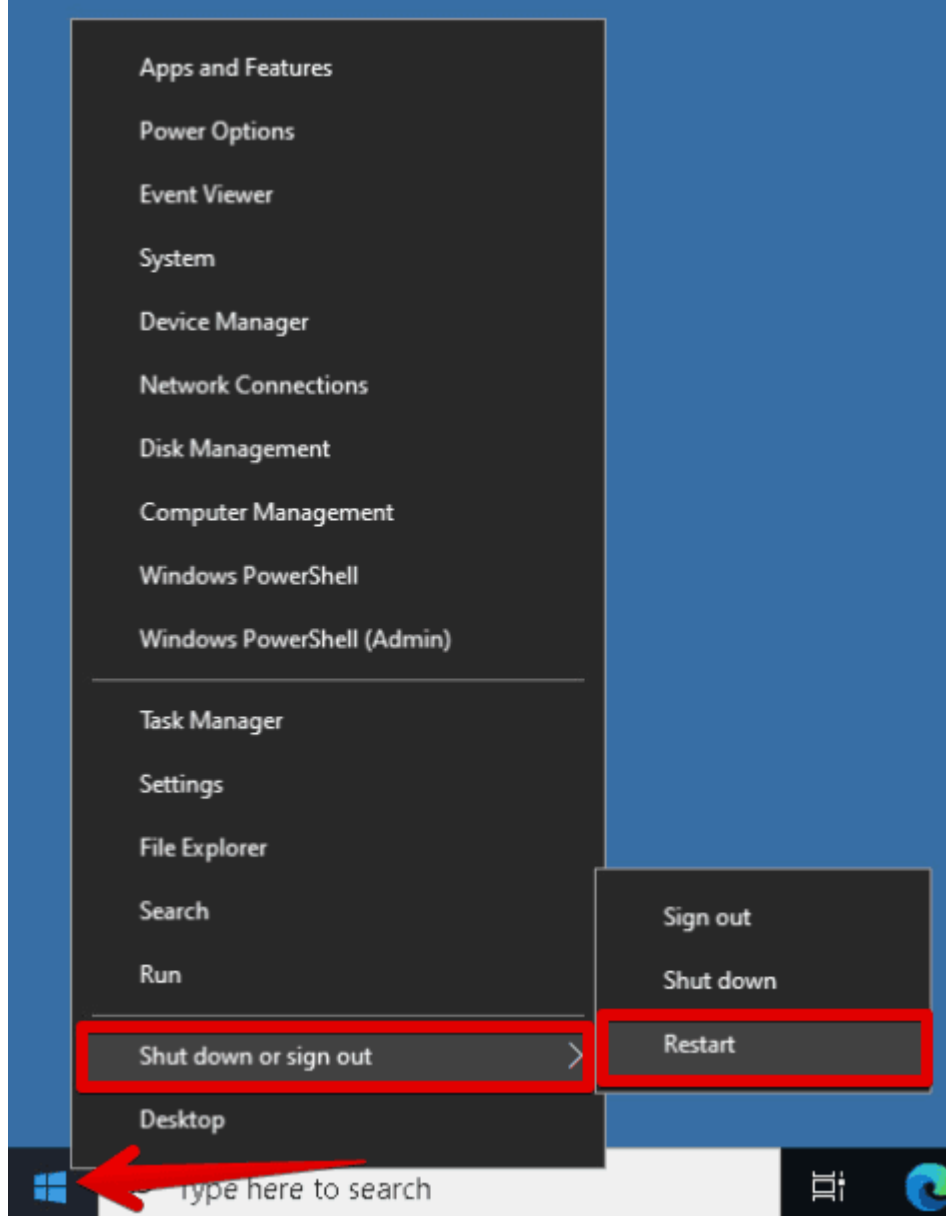
Click OK.



Note: Repeat Steps 11-13 on ACIDC01.

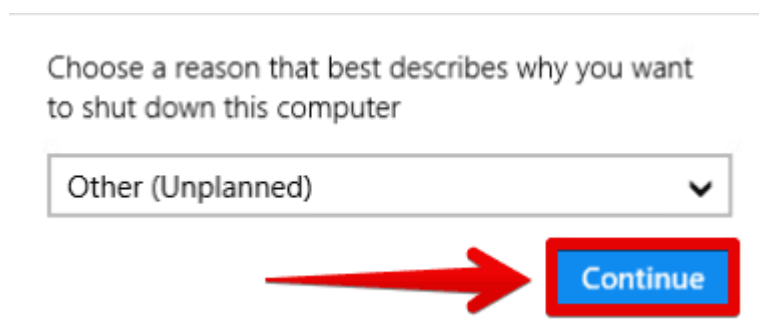
15

Right-click Start, select Shut down or sign out and click Restart.



16

In the pop-up window, **select** Other (Planned) in the drop-down menu and **click** Continue.



Note: The device must be restarted to ensure that the Default Group Policy applies successfully.



Alert: Ensure the device has restarted successfully before continuing with the next task. Reconnect to the device to ensure it has restarted successfully.

Task 2 – Generate Logs for Evaluation

Logs must be generated to simulate a brute-force attack on a lab device. A script file has been created on the **ACIKALI** device to achieve this.

This task will simulate a brute-force attack on an SMB file share.

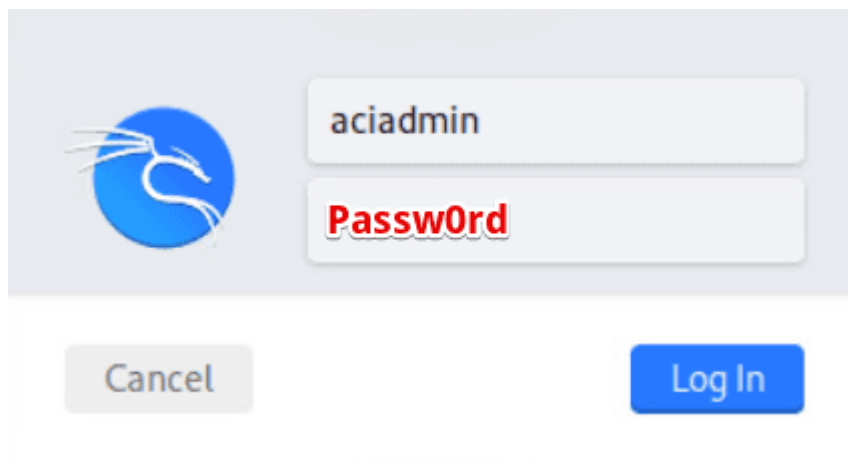
1

Connect to **ACIKALI** on the **sign-in screen**, **enter** the following **credentials**, and **press**

Enter:

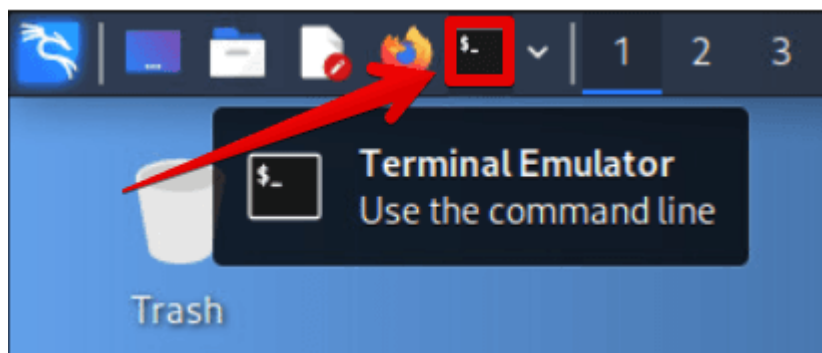
Username: `aciadmin`

Password: `Passw0rd`



2

Open the **Terminal Emulator** from the **Activities** pane.



3

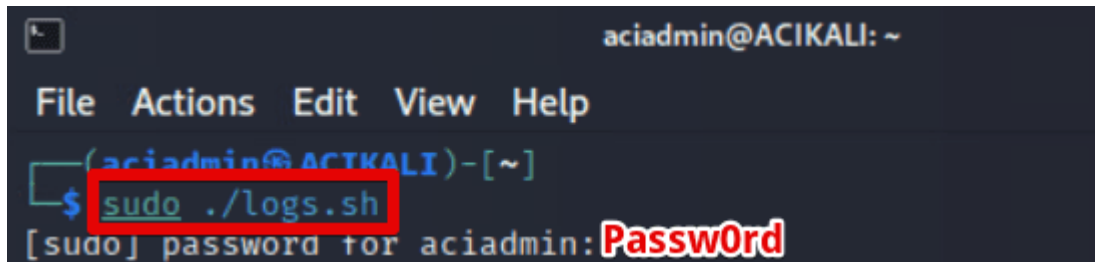
In the `aciadmin@ACIKALI:~` window, **type** the **following** and **press** Enter:

```
sudo ./logs.sh
```

Enter the following when prompted for a password:

Passw0rd

Press Enter.

A terminal window titled 'aciadmin@ACIKALI: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(aciadmin@ACIKALI)-[~]'. The command 'sudo ./logs.sh' is entered and highlighted with a red box. The next line shows the password prompt '[sudo] password for aciadmin:' followed by the password 'Passw0rd' in red text.

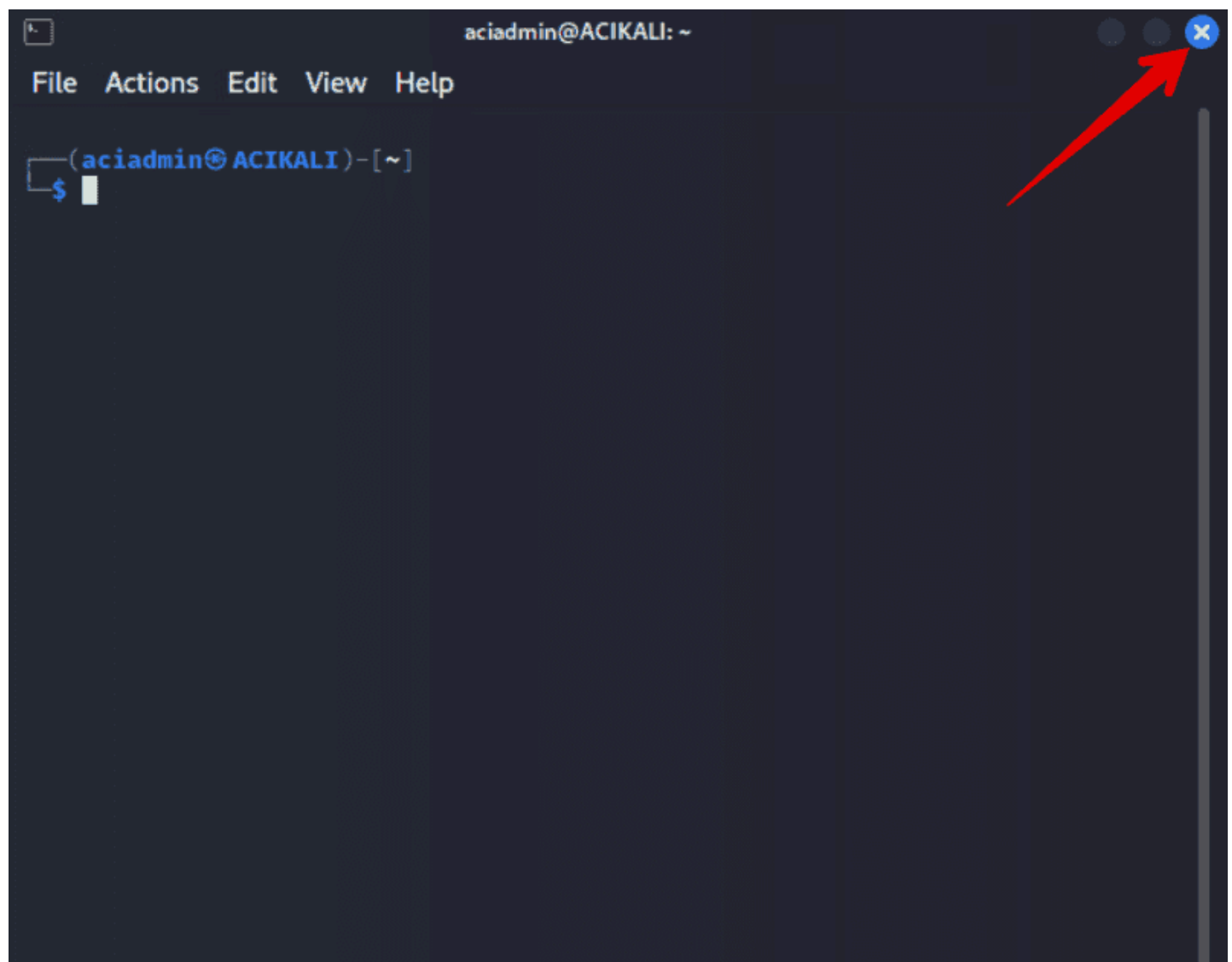
```
aciadmin@ACIKALI: ~  
File Actions Edit View Help  
(aciadmin@ACIKALI)-[~]  
$ sudo ./logs.sh  
[sudo] password for aciadmin: Passw0rd
```



Note: The script will take a couple of minutes to execute. There will be no output to the screen.

4

Close the aciadmin@ACIKALI:~ window.



Task 3 – Creating a Logon Events Script using PowerShell ISE

Failed logon events might indicate a brute-force attack on a device. A Cyber Security Analyst must monitor failed logon events to prevent these attacks.

In this task, a PowerShell Script will be created to monitor failed logon events on a remote device.

1

Connect to **ACIWIN11**, **click Start** and **type** the **following**:

```
PowerShell ISE
```

Press Enter.

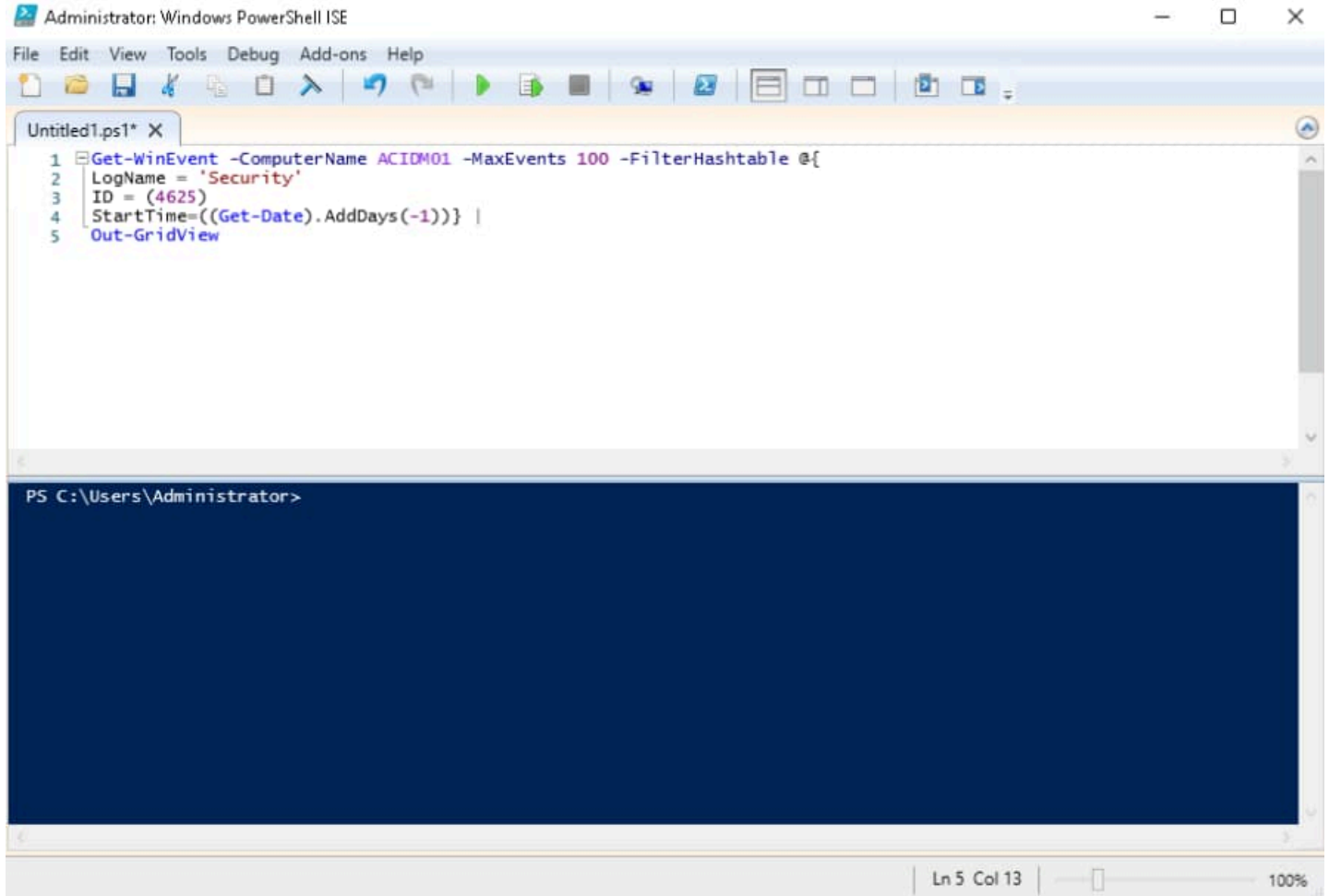
2

In the **Administrator: Windows PowerShell ISE** window, **click New Script**.

3

In the **Untitled.ps1** pane of the **Administrator: Windows PowerShell ISE** window, **enter** the **following**:

```
Get-WinEvent -ComputerName ACIDM01 -MaxEvents 100 -FilterHashtable @{  
LogName = 'Security'  
ID = (4625)  
StartTime=((Get-Date).AddDays(-1))} |  
Out-GridView
```



The screenshot shows the Windows PowerShell ISE interface. The title bar reads 'Administrator: Windows PowerShell ISE'. The menu bar includes 'File', 'Edit', 'View', 'Tools', 'Debug', 'Add-ons', and 'Help'. The toolbar contains various icons for file operations, editing, and execution. The script editor shows a file named 'Untitled1.ps1' with the following PowerShell script:

```
1 Get-WinEvent -ComputerName ACIDM01 -MaxEvents 100 -FilterHashtable @{
2   LogName = 'Security'
3   ID = (4625)
4   StartTime=((Get-Date).AddDays(-1))} |
5   Out-GridView
```

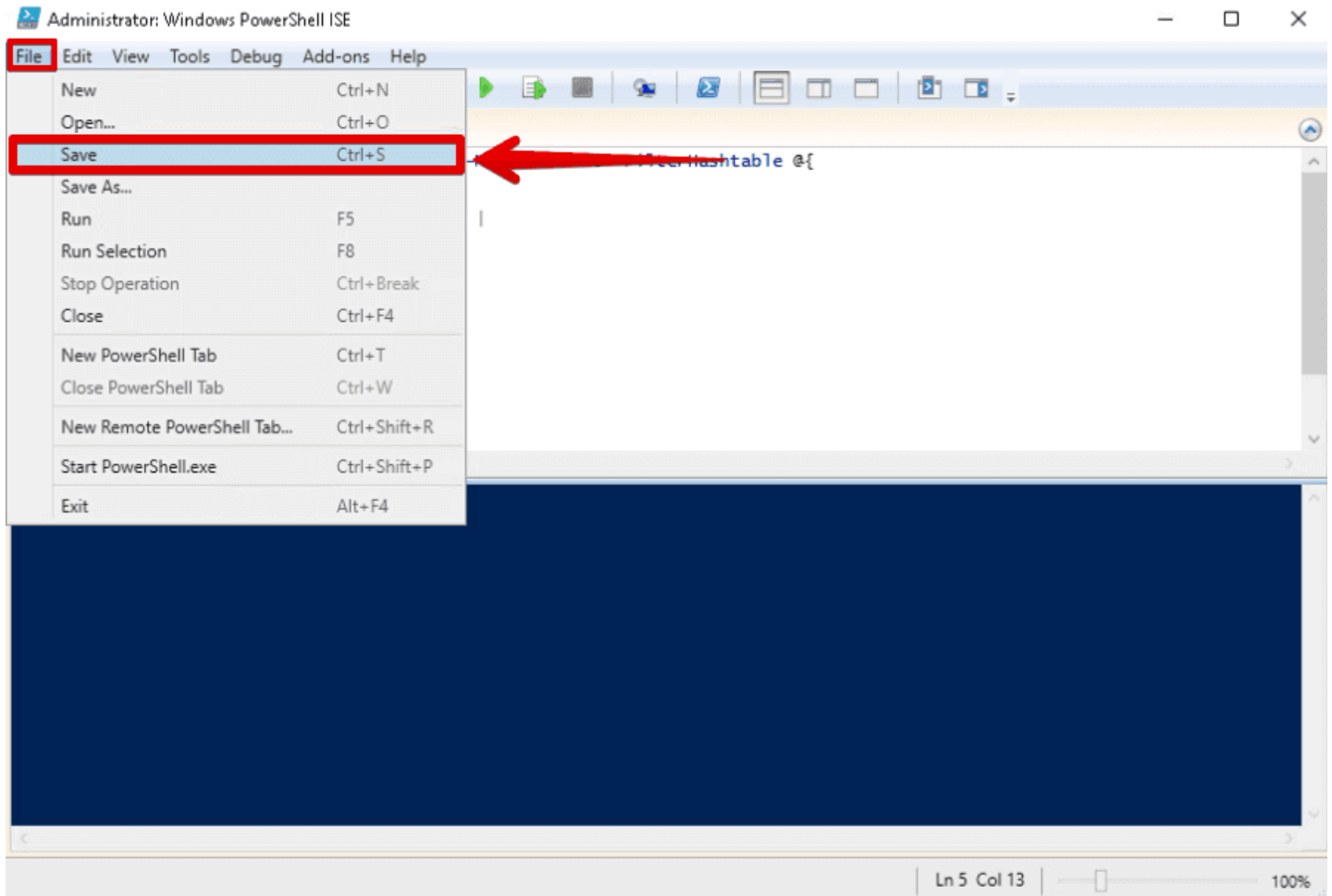
Below the script editor is a console window with the prompt 'PS C:\Users\Administrator>'. The status bar at the bottom indicates 'Ln 5 Col 13' and '100%' zoom.



Alert: It is essential to keep the script formatting as displayed to ensure functionality.

4

On the **Administrator: Windows PowerShell ISE** window, **select File** and **click Save**.



5 In the **Save As** window, **enter** the following in the **File name** field:

```
Monitor Failed Logons
```

Click **Save**.

6 **Close** the **Administrator: Windows PowerShell ISE** window.

7 **Right-click** **Start** and **select** **Windows Terminal (Admin)**.

8 In the **Administrator: Windows PowerShell** window, **type** the following:

```
Set-ExecutionPolicy Unrestricted
```

Press **Enter**.

9 In the **Administrator: Windows PowerShell** window, **type** the following:

```
cd .\Documents\
```

Press **Enter**.

10

In the **Administrator: Windows PowerShell** window, **type** the **following**:

```
& '.\Monitor Failed Logons.ps1'
```

Press **Enter**.

11

Close the **Out-GridView** window.



Note: For the output of the screen, it can be seen that there were several attempts to access the account. The created script displays the logs for the last day and the first 100 events. This is an indication of a brute-force attack on the account. The Event ID for successful logins is 4624, and 4625 for failed logins. Several other Event IDs are available that can be used to monitor events on a device.

Task 4 – Preventing a Brute-Force Attack

After determining that an attempt for a brute-force attack on an account, a Cyber Security Specialist needs to be able to implement measures to prevent these types of attacks.

In this task, a policy will be applied to prevent a brute-force attack on a domain account.

1

Connect to **ACIDC01** in **Server Manager**, **click** **Tools** and **select** **Group Policy Management**.

2

In the **Group Policy Management** window, **ensure** the **ACIPLAB.COM** is expanded and **select** the **Default Domain Policy**.

In the pop-up window, **click** **OK**.

3

Right-click the **Default Domain Policy** and **select** **Edit**.

4

Expand Computer Configuration >> Policies >> Windows Settings >> Security Settings
and **click** Account Policies.

5

In the Account Policies pane, **right-click** Account Lockout Policy and **select** Open.

6

Double-click the Account lockout threshold.

7

In the Account lockout threshold Properties window **change**, the Account will not lock
out after: value to the following:

5

Click OK.

8

In the Suggested Value Changes pop-up window, **click** OK.

Close all open windows.

9

Right-click Start and **select** Windows Powershell (Admin).

10

In the Administrator: Windows PowerShell window, **type** the following and **press** Enter:

```
gpupdate /force
```

11

Connect to ACIDM01, **right-click** Start, and **select** Windows Powershell (Admin).

12

In the **Administrator: Windows PowerShell** window, **type** the following and **press Enter**:

```
gpupdate /force
```

13

Connect to **ACIKALI** and **open** the **Terminal Emulator** from the **Activities** pane.

If the device is locked, **sign in** with these **credentials**:

Username: **aciadmin**

Password: **Passw0rd**

14

Enter the following in the **aciadmin@ACIKALI:~** window:

```
sudo ./logs.sh
```

Press Enter.

When prompted, **enter** the following password and **press Enter**:

```
Passw0rd
```

15

Reconnect to the **ACIWIN11** device, **click Start** and **type PowerShell ISE**.

Press Enter.

16

In the **Administrator: Windows PowerShell ISE** window, **click File** and **select Open**.

17

In the **Open** window, **select Monitor Failed Logins** script and **click Open**.

18

In the **Monitor Failed Logins** pane, **change** the following values:

```
ComputerName ACIDC01
```

```
ID = (4740)
```

19

Click **Run Script (F5)** in the **Administrator: Windows PowerShell ISE** window.

20

In the pop-up window, **click** **OK**.

21

Close the **Out-GridView** window once you have viewed the results.



Note: From the results, it can be seen that the account was locked after several failed login attempts, thus preventing a brute-force password attack. This indicated that the applied policy was successful in preventing the attack.

Exercise 2 – Monitor Login Events on Linux Devices

A Cyber Security Analyst must constantly monitor the networking environment for malicious activities. Failed login attempts might indicate a brute-force attack on the device.

In this exercise, the login attempts on a Linux device will be monitored.

Learning Outcomes

After completing this exercise, you should be able to:

- Monitor Login Events on a Linux Device

Your Devices

You will be using the following devices in this lab. Please power these on now.

- **ACIDC01** – Windows Server 2022 – Domain Controller
- **ACIDM01** – Windows Server 2022 – Domain Member Server
- **ACIALMA** – ALMA Linux Standalone Workstation

Task 1 – Monitor Login Events on a Linux Device

Monitoring the users that log into a device is essential to determine if only authorized users are accessing the device.

In this task, the users that accessed the device will be monitored.

1 **Connect** to **ACIALMA** from the **Activities window** and **click Terminal**.

2 In the **Terminal window**, **type** the following and **press Enter**:

```
sudo last -w
```

When prompted for a **password**, **enter** the **following**:

```
Passw0rd
```



Note: Executing the command displays the user's time and date that was logged into the system. The administrator can use this information to determine if unauthorized access to the device has been gained.

3 In the **Terminal window**, **type** the **following** and **press Enter**:

```
sudo last reboot
```



Note: Executing the command displays when the system was last restarted. This can determine if the server was rebooted without the administrator's knowledge.

4 In the **Terminal window**, **type** the **following** and **press Enter**:

```
sudo last -w -i
```



Note: Executing the command displays the user, time and IP address that was used to login into the server.

5 **Connect** to **ACIDM01**, **right-click** **Start** and **select** **Windows PowerShell (Admin)**.

6 In the **Administrator: Windows PowerShell** window, **type** the following:

```
ssh aciadmin@192.168.0.4
```

Type **yes** when prompted and **press** **Enter**.

When prompted for a **password**, **type** the following:

```
wrongpassword
```

Press **Enter**.

7 **Type** the following in the **Administrator: Windows PowerShell** window and **press** **enter**:

```
wrongpassword
```

8 In the **Administrator: Windows PowerShell**, **type** the following and **press** **Enter**:

```
wrongpasswordagain
```



Note: The connection is refused after the third attempt to connect to the **ACIALMA** device using an SSH connection.

9 **Connect** to **ACIALMA** and **ensure** the **Terminal window** is open.

Type the following:

```
sudo last -F
```

Press **Enter**.

10 **Reconnect** to **ACIDM01** and **ensure** the **Administrator: Powershell window** is open.

11 In the **Administrator: Windows Powershell** window, **type** the following and **press** Enter:

```
ssh aciadmin@192.168.0.4
```

Type the following when prompted for a password:

```
Passw0rd
```

12 **Reconnect** to **ACIALMA** and **ensure** the **Terminal window** is open.

13 In the **Terminal window**, **type** the following and **press** Enter:

```
sudo last -w
```



Note: Executing the command displays the users that are currently logged in. It can be seen that the **aciadmin** user is locally logged in and connected through an ssh session. A Cyber Security Analyst can monitor these logins to ensure only authorized users access devices on the network.

These commands were executed on the local system, but these logs can be aggregated on the remote device running a SIEM application where several devices can be monitored.

Keep all **devices** that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Techniques to Determine Malicious Activity** Practice Lab.

Summary

You completed the following exercises:

Exercise 1 – Monitor Windows Event Log with a PowerShell Script

Exercise 2 – Monitor Login Events on Linux Devices

You should now be able to:

- Enable Audit Logon Events
- Generate Logs for Evaluation

- Create a Logon Events Script using PowerShell ISE
- Prevent a Brute-Force Attack
- Monitor Login Events on a Linux Device

Feedback



Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.